1. RtHDDump Content Difference Report

1.1. Abstract

This report investigates RtHDDump file **content** differences and their relationship to device states (preferred device, headset plug state, and Dolby/enhancement modes). The analysis focuses on WID lines and registry-style key/value entries inside each dump rather than relying on filename tokens. We find that WID lines are identical across all dumps, while the state changes consistently align with differences in specific `REG_*` keys.

1.2. Data and methods

- Dataset: 27 RtHDDump captures.
- WID analysis: extract all lines starting with `Wid=` and compare across files.
- Content difference analysis: extract key/value lines (`<key> = <value>`).
- Paired comparison method: for each device state, match files with all other states held constant and compare their key/value lines. A key is a **signature** when it changes in ≥ half of the paired comparisons for that state.
- Values are trimmed for readability (`prefix…suffix`) but still show the changing portions.

1.3. Results

1.3.1. WID section stability

All 27 files contain the same 11 WID lines with identical values, indicating that the device-state verbs are **not** encoded by WID changes in this dataset.

WID	Codec	Drv	Loc
12	40000000	40000000	00000000
13	411111F0	411111F0	00000000
14	90170120	90170120	00000000
17	90170120	90170120	00000000
18	81D111F0	—	—
19	03A11030	03A11030	00080000
1A	411111F0	411111F0	00020400
1B	411111F0	411111F0	00000000
1D	40471A6D	411111F0	00000000
1E	411111F0	411111F0	00000700
21	03211010	03211010	00020000

1.3.2. Feature difference summary

Feature	Paired comparisons	Keys ≥ half pairs	Keys all pairs
preferred_device	3	15	8
headset_plugged	1	36	36
speaker_dolby	6	5	3
speaker_enhance	6	9	3
headset_dolby	3	10	10
headset_enhance	4	14	3

preferred_device	15
headset_plugged	36
speaker_dolby	5
speaker_enhance	9
headset_dolby	10
headset_enhance	14

Table 1: Signature key counts by feature (≥ half of paired comparisons)

1.3.3. Preferred device signatures

Key	Coverage	Speaker preferred	Headset preferred
`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`	3/3	`02 00 00 00 01 00 00 00 EF 06`	`02 00 00 00 01 00 00 00 EF 01`
`(REG_BINARY) Position`	3/3	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4`	3/3	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…0 00 40 34 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00`
`(REG_SZ) {24dbb0fc-9311-4b3d-9cf0-18ff155639d3},3`	3/3	`{0.0.0.00000000}.{45e064d4-3174-4fd5-ace4-5287bc0bdf22}`	`{0.0.0.00000000}.{105abd99-be3c-4986-856c-3f24ec337b55}`
`(REG_BINARY) {bb8bdb4a-edac-4660-9056-8e67e68e4e77},4`	3/3	`41 00 00 00 01 00 00 00 C7 1B 04 75 36 2…3 E0 37 6D 08 5A`	`41 00 00 00 01 00 00 00 0B B9 7A 91 18 0…9 1A 56 F8 04 5A`

1.3.4. Headset plug signatures

Key	Coverage	Headset plugged	Headset unplugged
`(REG_DWORD) InternalSpeakerStreamActive`	1/1	`0x0000`	`0x0001`
`(REG_BINARY) {5510c7ab-dfc2-40d0-a98b-5f77f697005e},2`	1/1	`03 00 00 00 01 00 00 00 28 00 00 00`	`03 00 00 00 01 00 00 00 00 00 00 00`
`(REG_BINARY) {5510c7ab-dfc2-40d0-a98b-5f77f697005e},1`	1/1	`03 00 00 00 01 00 00 00 08 00 00 00`	`03 00 00 00 01 00 00 00 02 00 00 00`
`(REG_DWORD) {3ba0cd54-830f-4551-a6eb-f3eab68e3700},6`	1/1	`0x0000`	`0x0001`
`(REG_BINARY) {194ef948-7cdb-403e-9f47-19418f7b248d},3`	1/1	`40 00 00 00 01 00 00 00 8D F3 96 28 D4 9A DC 01`	`40 00 00 00 01 00 00 00 59 36 9F D1 D3 9A DC 01`

1.3.5. Speaker Dolby signatures

Key	Coverage	Speaker Dolby on	Speaker Dolby off

`(REG_BINARY) Position`	6/6	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4`	6/6	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 60 23 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1`	6/6	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 60 23 00 00`
`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`	4/6	`02 00 00 00 01 00 00 00 20 0C`	`02 00 00 00 01 00 00 00 EF 07`
`(REG_BINARY) {8a845654-d6c3-4cd7-b4eb-243d4bd99032},2`	3/6	`41 00 00 00 01 00 00 00 00 00 00 00 00 0…E F9 95 53 37 90`	`41 00 00 00 01 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`

1.3.6. Speaker enhancement signatures

Key	Coverage	Speaker enhancement on	Speaker enhancement off
`(REG_BINARY) Position`	6/6	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4`	6/6	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1`	6/6	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 23 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00`
`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`	5/6	`02 00 00 00 01 00 00 00 EF 01`	`02 00 00 00 01 00 00 00 EF 02`
`(REG_DWORD) {1da5d803-d492-4edd-8c23-e0c0ffee7f0e},5`	4/6	`∅`	`0x0001`

1.3.7. Headset Dolby signatures

Key	Coverage	Headset Dolby on	Headset Dolby off
`(REG_BINARY) {8a845654-d6c3-4cd7-b4eb-243d4bd99032},2`	3/3	`41 00 00 00 01 00 00 00 00 00 00 00 00 0…F E8 0F 4D 39 5D`	`41 00 00 00 01 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`
`(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2`	3/3	`02 00 00 00 01 00 00 00 EF 07`	`02 00 00 00 01 00 00 00 EF 05`
`(REG_BINARY) Position`	3/3	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`	`01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00`
`(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4`	3/3	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…0 00 80 34 00 00`	`41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 C0 33 00 00`

(REG_BINARY) {6737016f-5360-48ee-af05-e616c8ff27a7},2	3/3	02 00 00 00 01 00 00 00 04 00	02 00 00 00 01 00 00 00 00 00

1.3.8. Headset enhancement signatures

Key	Coverage	Headset enhancement on	Headset enhancement off
(REG_BINARY) Position	4/4	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00	01 00 00 00 00 00 00 00 00 00 00 00 00 0…0 00 00 00 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},4	4/4	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 24 00 00
(REG_BINARY) {1b4dab55-b1fb-4d8c-8317-f2d4a96efbb8},1	4/4	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 A0 23 00 00	41 00 00 00 01 00 00 00 01 00 01 00 01 0…2 00 20 24 00 00
(REG_BINARY) {1e94c58f-3e40-4ddb-b10c-a86d8b870a31},2	3/4	02 00 00 00 01 00 00 00 EF 01	02 00 00 00 01 00 00 00 EF 02
(REG_BINARY) {624f56de-fd24-473e-814a-de40aacaed16},3	2/4	41 00 00 00 01 00 00 00 FE FF 02 00 80 B…0 AA 00 38 9B 71	∅

1.4. Conclusion

The device-state verbs are reflected in **registry key/value differences**, not WID line changes. Preferred device and headset plug state show strong, consistent content signatures, while Dolby and enhancement states show fewer but still repeatable key changes across paired comparisons. The signature tables above provide the definitive, content-based mapping from each state to the specific keys that change.

1.5. Appendix: Regeneration

Generate the CSV summary and the PDF report from the current dumps:

```
python rthddump_report.py --output rthddump_summary.csv
typst compile RtHDDump_Report.typ RtHDDump_Report.pdf
```